Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits
CSE111	ORIENTATION TO COMPUTING-I	2	0	0	2
Course Weightage	ATT: 20 CA: 50 ETT: 30	**Exam Category: X1: Mid Term Exam: Not Applicable – End Term Exam: All MCQ**			
Course Focus	SKILL DEVELOPMENT,EMPLOYABILITY				

Course Outcomes :Through this course students should be able to

CO1 :: understand the various functional components of a computer system and basics of computer language.

CO2 :: explain operating system components and functionalities, and manage various file systems and processes in Windows and Linux.

CO3 :: describe Linux OS features, installation, directory structure, disk partitions, shell commands, kernel types, and comparison with Windows OS.

CO4 :: predict cohorts based on their technical skillset, understand the relevance of various pathways, and obtain insights of MOOC 's platforms.

CO5 :: understand network components, configurations, and server types, and identify and mitigate security issues effectively.

CO6 :: practice technical concepts of version control using git and GitHub and create technical profiles on different computing platforms

	Reference Books (R)		
Sr No	**Title**	**Author**	**Publisher Name**
R-1	RED HAT RHSCA/RHCE 7	SANDER VAN VUGT	PEARSON
R-2	OPERATING SYSTEM CONCEPTS	ABRAHAM SILBERSCHATZ, PETER B. GALVIN, GERG GAGNE	WILEY

Relevant Websites (RW)		
Sr No	**(Web address) (only if relevant to the course)**	**Salient Features**
RW-1	https://www.mysmartprice.com/gear/laptop-pc-model-system-configuration-details-check/	System Configuration
RW-2	https://www.geeksforgeeks.org/what-are-threads-in-computer-processor-or-cpu/#:~:text=Threads%20are%20the%20virtual%20components,it%20will%20have%204%20threads	Example of Smartphone application
RW-3	https://www.hp.com/us-en/shop/tech-takes/how-to-enter-bios-setup-windows-pcs	BIOS Setup on Windows PCs
RW-4	https://support.microsoft.com/en-us/windows/pair-a-bluetooth-device-in-windows-2be7b51f-6ae9-b757-a3b9-95ee40c3e242#:~:text=On%20your%20PC%2C%20select%20Start%20%3E%20Settings%20%3E%20Devices%20%3E%20Bluetooth,they%20appear%2C%20then%20select%20Done.	Pairing Bluetooth device to a laptop

RW-5	https://careerservices.fas.harvard.edu/channels/create-a-resume-cv-or-cover-letter/	Profile creation and Good CV Practices

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Computer Systems(Basic structure of computer and its working, Computer associated peripherals, Memories - RAM, ROM, Secondary storage devices, System Configuration – features and comparison (SSD vs hybrid, types of RAMs, Processors - cores/threads))	R-1 R-2	RW-1 RW-2	L1: Lecture zero to be delivered so that students will understand why they need to study this subject. L2: Discuss the basic structure of a computer system along with its peripheral devices.	Students would be able to understand the basic structure of a computer system along with its peripheral devices.	Demonstration with PowerPoint presentation	Demonstration of various peripheral devices (like the use of a mouse, keyboard, etc.) could be delivered.
	Lecture 2	Computer Systems(BIOS Configuration, Compare and contrast PC connection interface (USB, SATA, HDMI, NFC, Bluetooth), RAID, GPU basics, Synchronization across CPU and GPU.)	R-2	RW-3 RW-4	Discuss the BIOS configuration and explain the various PC connection interfaces.	Students will be able to know about the various PC connection interfaces	Demonstration with PowerPoint presentation	The use of USB and HDMI cables with the computer system.
	Lecture 3	Computer Languages (Machine language, Assembly language, High level language, Steps in development of a program, Compilation and Execution, Compiler, Interpreter, Assembler.)	R-1 R-2		Discuss the types of technical languages along with the compiler, interpreter, and assembler.	Students will be able to understand the types of technical languages along with the compiler, interpreter, and assembler.	Demonstration with PowerPoint presentation	Use of compiler for executing a program.

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 2	Lecture 4	Operating System(Operating Systems and its components, Windows Operating System Versions and features, Installation Process)	R-2		Discuss the different versions of the Windows Operating System.	Students will understand the different versions of Windows Operating System.	Demonstration with PowerPoint presentation	Live installation of any Windows OS
	Lecture 5	Operating System(Directory Hierarchy of Windows Operating System (Single level and multiple level), Bootloader.)	R-2		Discuss the Directory Hierarchy of the Windows Operating System.	Students will learn about the directory hierarchy of the Windows operating system.	Demonstration with PowerPoint presentation	
	Lecture 6	File system management (File system basics, Types of file systems (FAT, GFT, HFS, NDFS, UDF, Extended file systems))	R-2		Discuss the different types of file systems like FAT, GFT, HFS, etc.	Students will learn about the usability of different types of file systems.	Demonstration with PowerPoint presentation	
Week 3	Lecture 7	File system management (Pipes and redirection, Searching the file system using find and grep with simple regular expressions, Basic process control using signals)	R-2		Discuss the searching operations for different file systems like FAT, GFT, HFS, etc.	Students will be able to learn about the search operations for different types of file systems.	Demonstration with PowerPoint presentation	
	Lecture 8	File system management (Pausing and Resuming process from a Linux terminal, terminating a process, Adding/removing from search path using PATH variable.)	R-2		Discuss the mechanism of pausing, resuming, and termination of a process from a Linux terminal.	Students will be able to know about the mechanism of pausing, resuming, and termination of a process from the Linux terminal.	Demonstration with PowerPoint presentation	
	Lecture 9	Linux Operating System (Linux OS and its features, Distribution versions, installation process, Directory Hierarchy of Linux System (single level and multiple level).)	R-2		Discuss the Linux OS and its directory hierarchy.	Students will be able to understand the Linux OS and its directory hierarchy.	Demonstration with PowerPoint presentation	
Week 4	Lecture 10				Test			
	Lecture 11	Linux Operating System (Partitions: Understanding disk partitions and obtaining partition information using system tools, Comparison of windows and Linux OS, Virtual Machines.)	R-2		Discuss the disk partitions using system tools. Also, compare Windows and Linux OS.	Students will learn about disk partitioning and the difference between Windows and Linux OS.	Demonstration with PowerPoint presentation	

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 4	Lecture 12	Other Shell commands(ls, cat, man, cd, touch, cp, mv, rmdir, mkdir, rm, chmod)	R-1 R-2			Discuss the execution of different types of Unix commands.	Students will get to know about the execution of different types of Unix commands.	Demonstration with PowerPoint presentation	Live execution of commands using terminal
Week 5	Lecture 13	Other Shell commands(pwd, ps, kill, etc, Kernel and types of kernels.)	R-1 R-2			Discuss about the execution of different types of shell commands.	Students will get to know about the execution of different types of shell commands.	Demonstration with PowerPoint presentation	Live execution of commands using terminal
		SPILL OVER							
Week 5	Lecture 14					Spill Over			
		MID-TERM							
Week 5	Lecture 15	Cohorts and Skill Sets (Introduction to Cohorts)	R-1			Discuss about the cohorts and its significance	Students would be able to understand the different types of cohorts and their significance	Demonstration with PowerPoint presentation.	
		Cohorts and Skill Sets (Purpose of Cohorts)	R-1			Discuss about the cohorts and its significance	Students would be able to understand the different types of cohorts and their significance	Demonstration with PowerPoint presentation.	
		Cohorts and Skill Sets (Companies)	R-1			Discuss about the cohorts and its significance	Students would be able to understand the different types of cohorts and their significance	Demonstration with PowerPoint presentation.	
		Cohorts and Skill Sets(Skills required and skill sources for different Cohorts (Internal and External))	R-1			Discuss about the cohorts and its significance	Students would be able to understand the different types of cohorts and their significance	Demonstration with PowerPoint presentation.	
Week 6	Lecture 16	Types of Cohorts(Cloud Computing)	R-1			Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	

Week 6	Lecture 16	Types of Cohorts(Cyber Security)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	
		Types of Cohorts(Data Science)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	
		Types of Cohorts(Full Stack Development)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	
		Types of Cohorts(Machine Learning)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	
	Lecture 17	Types of Cohorts(Software Methodologies and Testing)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 6	Lecture 17	Types of Cohorts(UI/UX)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	
		Types of Cohorts(Metaverse and Internet of Things)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	
		Types of Cohorts(Job Roles for Different Cohorts)	R-1		Discuss about the various types of cohorts (like cloud, cyber security, networks etc.) and their importance in IT sector.	Students would be able to get the exposure about the various types of cohorts and accordingly they can set their skill set based targets for the placements.	Demonstration with PowerPoint presentation	
	Lecture 18				Certification - MOOCs			
Week 7	Lecture 19	Pathways(Introduction to Pathways)	R-1 R-2		Discuss the various types of pathways available for students	Students will be able to understand about the different pathways.	Demonstration with PowerPoint presentation	
		Pathways(Purpose of Pathways)	R-1 R-2		Discuss the various types of pathways available for students	Students will be able to understand about the different pathways.	Demonstration with PowerPoint presentation	
		Pathways(Job Roles for Different Pathways)	R-1 R-2		Discuss the various types of pathways available for students	Students will be able to understand about the different pathways.	Demonstration with PowerPoint presentation	
	Lecture 20	Pathways(Types of Pathways: Product Based, Service Based, Government Jobs, Higher studies, Entrepreneurship)	R-1 R-2		Discuss about the various types of pathways available for students	Students will be able to understand about the different pathways.	Demonstration with PowerPoint presentation	

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 7	Lecture 21	MOOCs and Hackathons (Introduction to MOOCs and Hackathons)	R-1 R-2		Discuss about the different types of MOOC and Hackathons available nowadays. Also, explain the importance of doing MOOCs and participating in hackathons.	Students will learn about the importance of doing MOOCs and appearing in hackathons for their professional growth.	Demonstration with PowerPoint presentation	Figma, GitHub, Stack Overflow, HackerRank, HackerEarth, GeeksforGeeks.
		MOOCs and Hackathons (Types of MOOCs)	R-1 R-2		Discuss about the different types of MOOC and Hackathons available nowadays. Also, explain the importance of doing MOOCs and participating in hackathons.	Students will learn about the importance of doing MOOCs and appearing in hackathons for their professional growth.	Demonstration with PowerPoint presentation	Figma, GitHub, Stack Overflow, HackerRank, HackerEarth, GeeksforGeeks.
		MOOCs and Hackathons (Various MOOCs Platforms)	R-1 R-2		Discuss about the different types of MOOC and Hackathons available nowadays. Also, explain the importance of doing MOOCs and participating in hackathons.	Students will learn about the importance of doing MOOCs and appearing in hackathons for their professional growth.	Demonstration with PowerPoint presentation	Figma, GitHub, Stack Overflow, HackerRank, HackerEarth, GeeksforGeeks.
		MOOCs and Hackathons (Benefits of MOOCs)	R-1 R-2		Discuss about the different types of MOOC and Hackathons available nowadays. Also, explain the importance of doing MOOCs and participating in hackathons.	Students will learn about the importance of doing MOOCs and appearing in hackathons for their professional growth.	Demonstration with PowerPoint presentation	Figma, GitHub, Stack Overflow, HackerRank, HackerEarth, GeeksforGeeks.

Week 7	Lecture 21	MOOCs and Hackathons (Globally Recognized Hackathons and Competitions)	R-1 R-2		Discuss about the different types of MOOC and Hackathons available nowadays. Also, explain the importance of doing MOOCs and participating in hackathons.	Students will learn about the importance of doing MOOCs and appearing in hackathons for their professional growth.	Demonstration with PowerPoint presentation	Figma, GitHub, Stack Overflow, HackerRank, HackerEarth, GeeksforGeeks.
		MOOCs and Hackathons (MAANG Companies)	R-1 R-2		Discuss about the different types of MOOC and Hackathons available nowadays. Also, explain the importance of doing MOOCs and participating in hackathons.	Students will learn about the importance of doing MOOCs and appearing in hackathons for their professional growth.	Demonstration with PowerPoint presentation	Figma, GitHub, Stack Overflow, HackerRank, HackerEarth, GeeksforGeeks.
Week 8	Lecture 22	Computer Network and Communication(Network types (wired and wireless), Network topologies, Network communication devices (Routers, Switches, Modems, Hubs, access point), Setting IP addresses, sharing files and folders, Remote Login, SSH, Wireless Security (http vs https))	R-1 R-2		Discuss the various concepts of networking and the different topologies used with different devices	Students will learn the basic concepts of networking including topologies and networking devices	Demonstration with PowerPoint presentation	
	Lecture 23				Portfolio			
	Lecture 24	Computer Network and Communication(Client Server model, Types of Servers (Proxy servers, Application server, Web server, File server, Database server, Synchronization server, Log server).)	R-1 R-2		Discuss the types of servers that are used and the how essential is the client-server model. Also, provide an introduction to the basic security essentials that are used in day-to-day life.	Students will learn the workings of the client-server model along with the different terminologies used in the security.	Demonstration with PowerPoint presentation	

Week 9	Lecture 25	Security Essentials(Basic security threats (malwares, Phishing, Social engineering, Password cracking))	R-1 R-2		Discuss the types of servers that are used and the how essential is the client-server model. Also, provide an introduction to the basic security essentials that are used in day-to-day life.	Students will learn the workings of the client-server model along with the different terminologies used in the security.	Demonstration with PowerPoint presentation	
		Security Essentials (Password management (Password complexity, Change default passwords), Open WiFi vs. secure WiFi, Multi Factor authentication, Admin vs. user vs. guest Account.)	R-1 R-2		Discuss the types of servers that are used and the how essential is the client-server model. Also, provide an introduction to the basic security essentials that are used in day-to-day life.	Students will learn the workings of the client-server model along with the different terminologies used in the security.	Demonstration with PowerPoint presentation	
	Lecture 26	Version Control(Overview of Git and GitHub, Install git and create a GitHub account)	R-1 R-2		Discuss the process of creating an account on GitHub and installing Git.	Students will learn about the process of creating an account on GitHub and installing Git.	Demonstration with PowerPoint presentation	
		Version Control(Create a local git repository, Add a new file to the repository, Creating a commit, Creation of a new Branch)	R-1 R-2		Discuss the process of creating an account on GitHub and installing Git.	Students will learn about the process of creating an account on GitHub and installing Git.	Demonstration with PowerPoint presentation	
	Lecture 27	Profile Creation(Figma, GitHub, Stack overflow, HackerRank, HackerEarth, GeeksforGeeks.)		RW-5	Discuss the applications of Computer Science and Engineering. Also, provide insights to the students on how to write an impact full CV. Discuss about the different types of MOOC and Hackathons available nowadays. Also, explain the importance of doing MOOC and participating in hackathons.	Students will learn about the importance of doing MOOCs and appearing in hackathons for their professional growth.	Demonstration with PowerPoint presentation	
		SPILL OVER						
Week 10	Lecture 28				Spill Over			

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 10	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0404 (4 out of 4)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Certification - MOOCs	NO	50	CO3, CO4
Portfolio	Yes	50	CO1, CO2, CO3, CO4, CO5, CO6
Test	NO	50	CO1, CO2

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Certification - MOOCs	To aware students about the importance of non-technical assets to streamline their career along with the learning of technical skill set.	Non-Technical Online MOOC course: Students will complete one non-technical MOOC course (Minimum duration of 5 hr) and submit the completion certificate. Rubrics to be followed: Certificate Submission – 15 Marks Viva/Presentation – 15 Marks	Individual	Online	30	6 / 9
Portfolio	To aware the students about the availability of online technical platforms and nurture them for the preparation of Placement based activities	The portfolio needs to be prepared which includes CV along with technical MooC registration. Students will create their technical profiles and taking insights by doing some technical activities thereon. Various technical platforms like Github, Stack overflow, Hackerrank, Figma, Hackerearth, Geeks for Geeks and Linked in could be considered. Rubrics to be followed: Profile creation and technical activities done by student on various technical platforms [10 marks] CV with one technical MOOC course registration proof [10 marks] GitHub profile with repositories and forks [10 marks]	Individual	Online	30	4 / 12

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Test	To evaluate the conceptual and technical understanding about the various fundamentals concepts of computer system	Test 1 will be MCQ-based, containing 30 questions of 1 mark each, which includes analytical and logical scenario-based questions. Students would be evaluated based on their performance in the class test	Individual	Online	30	3 / 5

MOOCs/ Certification etc. mapped with the Academic Task(s)

Academic Task	Name Of Certification/Online Course/Test/Competition mapped	Type	Offered By Organisation
Certification - MOOCs	LINUX COMMANDS AND SHELL SCRIPTING	MOOCs	SKILLERA
Test	INTRODUCTION TO GIT AND GITHUB	MOOCs	COURSERA

Where MOOCs/ Certification etc. are mapped with Academic Tasks:
1. Students have choice to appear for Academic Task or MOOCs etc.
2. The student may appear for both, In this case best obtained marks will be considered.

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.